Explicit Transient MMS Report
`testHighOrderHeatEquationExplicit`

Automatically generated from current solver and results

April 23, 2026

1 Objective

This report documents solver `testHighOrderHeatEquationExplicit` and the current results under `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquationExplicit/result` It covers: governing equation, high-order spatial discretization, explicit time integration, error definitions, convergence-rate calculations, and per-example result summaries.

2 Governing Equation

The solver integrates the transient diffusion equation:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (\alpha \nabla T) + Q, \tag{1}$$

with constant material coefficients ($\alpha = 1$, $\rho = 1$, $C_p = 1$ in the current case setup). In code, the spatial residual is defined as $R = \nabla \cdot (\alpha \nabla T) + Q$, and the explicit update is:

$$T_i^{n+1} = T_i^n + \frac{\Delta t}{\rho C_p} R_i^n. \tag{2}$$

3 MMS Implemented in the Solver

The solver supports five manufactured-solution families. The implemented expressions are:

Table 1: MMS families implemented in `testHighOrderHeatEquationExplicit.C`.

Case	T_{ex}	Q
`example_0`	$T_{\text{ex}} = A\sin(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)\sin(\beta\pi t)$	$Q = A\sin(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)\left(\rho C_p\beta\pi\cos(\beta\pi t) + 3\alpha(\beta\pi)^2\sin(\beta\pi t)\right)$
`example_1`	$T_{\text{ex}} = A\cos(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)\sin(\beta\pi t)$	$Q = A\cos(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)\left(\rho C_p\beta\pi\cos(\beta\pi t) + 3\alpha(\beta\pi)^2\sin(\beta\pi t)\right)$
`example_2`	$T_{\text{ex}} = A\cos(\beta\pi x)\cos(\beta\pi y)\sin(\beta\pi z)\sin(\beta\pi t)$	$Q = A\cos(\beta\pi x)\cos(\beta\pi y)\sin(\beta\pi z)\left(\rho C_p\beta\pi\cos(\beta\pi t) + 3\alpha(\beta\pi)^2\sin(\beta\pi t)\right)$
`example_3`	$T_{\text{ex}} = A\cos(\beta\pi x)\cos(\beta\pi y)\cos(\beta\pi z)\sin(\beta\pi t)$	$Q = A\cos(\beta\pi x)\cos(\beta\pi y)\cos(\beta\pi z)\left(\rho C_p\beta\pi\cos(\beta\pi t) + 3\alpha(\beta\pi)^2\sin(\beta\pi t)\right)$
`example_4`	$T_{\text{ex}} = A\sin(kx)\sin(ky)\sin(kz)\sin(\omega\pi t),\ k = \frac{1}{2}\gamma\pi$	$Q = \rho C_p\,\partial_t T_{\text{ex}} - \alpha\nabla^2 T_{\text{ex}} = A\sin(kx)\sin(ky)\sin(kz)\left(\rho C_p\omega\pi\cos(\omega\pi t) + 3\alpha k\right.$

4 Numerical Method

4.1 Space: high-order LRE

When `useHighOrder_T=true`, diffusive face fluxes are integrated by face quadrature using LRE-reconstructed gradients (`gradScalarFaceQuad`, `faceQuadWeight`), and assembled with `fvc::div(fluxT_HO)`.

4.2 Time: explicit Euler

No implicit temperature linear solve is performed at each time step; integration is fully explicit at the cell level. In semi-discrete form:

$$\mathbf{M}\,\dot{\mathbf{T}} = \mathbf{R}(\mathbf{T}, t), \qquad \mathbf{M}_{ii} = \rho C_p V_i, \tag{3}$$

i.e. a diagonal (lumped) mass matrix naturally arising from cell-centered FV. A consistent non-diagonal mass matrix is not used here.

4.3 Boundary stencils

In `createFields.H`, `includePatchInStencils` enables boundary contribution for `fixedValue`, `fixedVoltage`, and `zeroGradient` patches.

5 Errors and Convergence Orders

5.1 Cell-centered errors

With $e_i = T_i - T_{\text{ex},i}$:

$$L_1 = \frac{\sum_i |e_i| V_i}{\sum_i V_i}, \tag{4}$$

$$L_2 = \sqrt{\frac{\sum_i e_i^2 V_i}{\sum_i V_i}}, \tag{5}$$

$$L_\infty = \max_i |e_i|, \tag{6}$$

$$\texttt{error\_T} = \sqrt{\sum_i e_i^2}. \tag{7}$$

5.2 Gauss-point reconstructed errors

The solver reconstructs T_h at cell quadrature points (using gradient/Hessian/third derivative according to LRE order) and computes G_1, G_2, G_∞ analogously.

5.3 Spatial and temporal rates

For each fixed Δt, spatial rates are estimated from fits $e \sim h^p$ (reported in `spatial_convergence_rate_dt_..._to` and pairwise slopes between consecutive meshes (reported in `..._pairwise.dat`). Temporal rates are estimated from finest-mesh errors ($N = 30$) via $e \sim (\Delta t)^{p_t}$.

6 Observed Run Configuration

- Domain: cube $[0, 1]^3$, uniform meshes $N = 10, 15, 20, 25, 30$.
- `endTime = 0.125` (from transient summaries).
- `DT_VALUES` in `run_convergence.py`: $[5 \times 10^{-3}, 10^{-3}, 5 \times 10^{-4}, 2.5 \times 10^{-4}, 10^{-4}, 10^{-5}]$.
- `useHighOrder_T=true`, with LRE settings `N=3, Nn=60, k=6, maxStencilSize=80, useQRDecomposition=` `useGlobalStencils=true`.

7 Results: folder `example_0`

Stored mmsType: `example_0`.

Exact temperature used: $T_{\text{ex}} = A\sin(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)\sin(\beta\pi t)$ **Source term used:** $Q = A\sin(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)\left(\rho C_p\beta\pi\cos(\beta\pi t) + 3\alpha(\beta\pi)^2\sin(\beta\pi t)\right)$

Stored parameters (from `exactSolutionProperties.used`): $A = 1.0$, $\beta = 4.0$, $\gamma = 3.0$, $\omega = 4.0$.

Table 2: Available Δt values in `example_0` (descending order).

Δt
5.000000e-03
1.000000e-03
5.000000e-04
2.500000e-04
1.000000e-04
1.000000e-05

Table 3: Global spatial fit rates p_{fit} (cell-centered norms) for `example_0`.

Δt	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	Comment
5.000000e-03	-52.182	-52.897	-53.399	-54.397	unstable/divergent
1.000000e-03	-205.545	-206.990	-207.970	-208.490	unstable/divergent
5.000000e-04	-178.351	-179.653	-180.555	-181.153	unstable/divergent
2.500000e-04	3.735	3.960	4.593	2.460	stable-like
1.000000e-04	3.739	3.964	4.594	2.464	stable-like
1.000000e-05	3.741	3.967	4.594	2.467	stable-like

Table 4: Global spatial fit rates p_{fit} (Gauss-point norms) for `example_0`.

Δt	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	-52.158	-52.970	-53.468
1.000000e-03	-205.358	-207.132	-208.977
5.000000e-04	-178.181	-179.795	-181.494
2.500000e-04	3.677	3.866	3.819
1.000000e-04	3.680	3.869	3.819
1.000000e-05	3.682	3.871	3.819

Short interpretation. Large time steps show explosive error growth (very large negative spatial slopes), consistent with explicit-scheme instability for diffusion-dominated dynamics. Within the stable subset, p_{L_2} ranges from 3.960 to 3.967, and p_{G_2} from 3.866 to 3.871.

8 Results: folder `example_1`

Stored mmsType: `example_1`.

Exact temperature used: $T_{\text{ex}} = A\cos(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)\sin(\beta\pi t)$ **Source term used:** $Q = A\cos(\beta\pi x)\sin(\beta\pi y)\sin(\beta\pi z)\left(\rho C_p\beta\pi\cos(\beta\pi t) + 3\alpha(\beta\pi)^2\sin(\beta\pi t)\right)$

Stored parameters (from `exactSolutionProperties.used`): $A = 1.0$, $\beta = 4.0$, $\gamma = 3.0$, $\omega = 4.0$.

Table 5: Final errors on the finest mesh (typically $N = 30$) for `example_0`.

Δt	N	L_2	G_2	L_∞	G_∞	$error_T$	final $\|R\|_\infty$	steps	t_{end}
5.000000e-03	30	7.180828e+31	9.830636e+31	9.616567e+32	3.144975e+33	1.179930e+34	6.909668e+36	25	0.12500
1.000000e-03	30	2.001368e+92	2.807602e+92	2.991551e+93	1.009176e+94	3.288583e+94	2.167778e+97	125	0.12500
5.000000e-04	30	8.427804e+97	1.187557e+98	1.104694e+99	3.776955e+99	1.384830e+100	8.044271e+102	250	0.12500
2.500000e-04	30	1.584127e-03	2.134367e-03	5.013583e-03	1.475396e-02	2.602987e-01	1.073363e-02	500	0.12500
1.000000e-04	30	1.575267e-03	2.126048e-03	5.012768e-03	1.475351e-02	2.588428e-01	1.105109e-03	1250	0.12500
1.000000e-05	30	1.569984e-03	2.121089e-03	5.012286e-03	1.475326e-02	2.579747e-01	6.561897e-03	12501	0.12501

Table 6: Temporal fit slopes from finest-mesh errors: $e \sim (\Delta t)^{p_t}$ for `example_0`.

Metric	p_t (all available Δt)	p_t (stable-like subset)
L1	27.329	0.002
L2	27.654	0.003
Linf	27.954	0.000
G1	27.284	0.002
G2	27.659	0.002
GInf	27.978	0.000

Table 7: Pairwise spatial convergence rates (all available Δt) for `example_0`.

Δt	N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	10	15	-57.451	-57.741	-57.121	-59.241	-57.403	-57.938	-57.213
5.000000e-03	15	20	-51.211	-52.427	-54.018	-53.927	-51.198	-52.395	-53.972
5.000000e-03	20	25	-48.768	-49.384	-49.976	-50.884	-48.775	-49.417	-50.124
5.000000e-03	25	30	-46.306	-47.061	-47.624	-48.561	-46.264	-47.142	-47.781
1.000000e-03	10	15	3.881	4.045	4.073	2.545	3.629	3.873	4.211
1.000000e-03	15	20	-303.121	-306.686	-308.931	-308.186	-302.294	-306.921	-312.183
1.000000e-03	20	25	-331.327	-332.187	-332.519	-333.687	-331.316	-332.225	-332.554
1.000000e-03	25	30	-294.297	-295.681	-297.354	-297.181	-294.272	-295.685	-297.363
5.000000e-04	10	15	3.883	4.047	4.074	2.547	3.630	3.875	4.211
5.000000e-04	15	20	3.396	3.910	5.321	2.410	3.587	3.847	5.158
5.000000e-04	20	25	-414.174	-420.565	-426.798	-422.065	-413.334	-420.843	-430.905
5.000000e-04	25	30	-748.083	-748.640	-748.135	-750.140	-748.073	-748.644	-748.107
2.500000e-04	10	15	3.884	4.048	4.074	2.548	3.631	3.875	4.212
2.500000e-04	15	20	3.400	3.914	5.323	2.414	3.590	3.850	5.158
2.500000e-04	20	25	3.937	3.910	4.993	2.410	3.798	3.861	1.705
2.500000e-04	25	30	3.833	3.928	3.279	2.428	3.815	3.892	3.132
1.000000e-04	10	15	3.884	4.049	4.075	2.549	3.632	3.876	4.212
1.000000e-04	15	20	3.402	3.917	5.324	2.417	3.591	3.852	5.157
1.000000e-04	20	25	3.943	3.917	5.002	2.417	3.803	3.866	1.705
1.000000e-04	25	30	3.847	3.944	3.266	2.444	3.825	3.903	3.132
1.000000e-05	10	15	3.885	4.050	4.075	2.550	3.632	3.876	4.212
1.000000e-05	15	20	3.403	3.919	5.324	2.419	3.593	3.853	5.157
1.000000e-05	20	25	3.947	3.921	5.007	2.421	3.805	3.869	1.705
1.000000e-05	25	30	3.856	3.953	3.258	2.453	3.831	3.909	3.132

Table 8: Available Δt values in `example_1` (descending order).

Δt
5.000000e-03
1.000000e-03
5.000000e-04
2.500000e-04
1.000000e-04
1.000000e-05

Table 9: Global spatial fit rates p_{fit} (cell-centered norms) for `example_1`.

Δt	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	Comment
5.000000e-03	-51.247	-52.234	-53.161	-53.734	unstable/divergent
1.000000e-03	-204.732	-206.819	-208.658	-208.319	unstable/divergent
5.000000e-04	-178.357	-180.243	-181.771	-181.743	unstable/divergent
2.500000e-04	3.797	3.957	4.361	2.457	stable-like
1.000000e-04	3.801	3.961	4.364	2.461	stable-like
1.000000e-05	3.804	3.963	4.366	2.463	stable-like

Table 10: Global spatial fit rates p_{fit} (Gauss-point norms) for `example_1`.

Δt	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	-51.250	-52.244	-53.176
1.000000e-03	-204.584	-206.968	-209.319
5.000000e-04	-178.223	-180.396	-182.447
2.500000e-04	3.719	3.865	3.965
1.000000e-04	3.722	3.868	3.964
1.000000e-05	3.724	3.870	3.964

Table 11: Final errors on the finest mesh (typically $N = 30$) for `example_1`.

Δt	N	L_2	G_2	L_∞	G_∞	$error_T$	final $\|R\|_\infty$	steps	t_{end}
5.000000e-03	30	4.150444e+31	5.764111e+31	9.615788e+32	3.167268e+33	6.819875e+33	6.918056e+36	25	0.12500
1.000000e-03	30	1.920615e+92	2.691570e+92	5.139745e+93	1.741483e+94	3.155892e+94	3.730646e+97	125	0.12500
5.000000e-04	30	1.138973e+98	1.616883e+98	2.561121e+99	8.879485e+99	1.871524e+100	1.871079e+103	250	0.12500
2.500000e-04	30	1.642459e-03	2.198700e-03	5.364449e-03	1.332935e-02	2.698836e-01	1.085436e-02	500	0.12500
1.000000e-04	30	1.633740e-03	2.190532e-03	5.343755e-03	1.332939e-02	2.684508e-01	1.255104e-03	1250	0.12500
1.000000e-05	30	1.628541e-03	2.185664e-03	5.331415e-03	1.332941e-02	2.675965e-01	6.619011e-03	12501	0.12501

Table 12: Temporal fit slopes from finest-mesh errors: $e \sim (\Delta t)^{p_t}$ for `example_1`.

Metric	p_t (all available Δt)	p_t (stable-like subset)
L1	27.144	0.002
L2	27.582	0.002
Linf	27.991	0.002
G1	27.102	0.002
G2	27.590	0.002
GInf	28.050	-0.000

Table 13: Pairwise spatial convergence rates (all available Δt) for `example_1`.

Δt	N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	10	15	-53.302	-54.276	-55.544	-55.776	-53.314	-54.173	-55.350
5.000000e-03	15	20	-53.934	-54.922	-55.524	-56.422	-53.964	-54.999	-55.681
5.000000e-03	20	25	-47.474	-48.436	-49.224	-49.936	-47.438	-48.501	-49.344
5.000000e-03	25	30	-45.254	-46.321	-47.433	-47.821	-45.231	-46.363	-47.459
1.000000e-03	10	15	3.743	3.962	4.577	2.462	3.582	3.795	2.676
1.000000e-03	15	20	-301.477	-307.095	-311.843	-308.595	-300.843	-307.409	-311.976
1.000000e-03	20	25	-329.090	-329.971	-331.737	-331.471	-329.092	-329.952	-331.655
1.000000e-03	25	30	-295.853	-297.201	-298.228	-298.701	-295.862	-297.170	-298.220
5.000000e-04	10	15	3.745	3.964	4.576	2.464	3.583	3.796	2.675
5.000000e-04	15	20	3.718	3.970	4.261	2.470	3.722	3.894	4.434
5.000000e-04	20	25	-416.811	-425.360	-432.865	-426.860	-415.993	-425.674	-433.280
5.000000e-04	25	30	-743.412	-744.196	-745.186	-745.696	-743.411	-744.201	-745.182
2.500000e-04	10	15	3.746	3.965	4.575	2.465	3.583	3.797	2.675
2.500000e-04	15	20	3.722	3.974	4.260	2.474	3.725	3.897	4.434
2.500000e-04	20	25	3.954	3.937	4.620	2.437	3.847	3.900	4.996
2.500000e-04	25	30	3.848	3.924	3.506	2.424	3.843	3.905	4.367
1.000000e-04	10	15	3.746	3.965	4.575	2.465	3.584	3.797	2.675
1.000000e-04	15	20	3.724	3.977	4.260	2.477	3.727	3.899	4.434
1.000000e-04	20	25	3.960	3.944	4.629	2.444	3.852	3.905	4.996
1.000000e-04	25	30	3.862	3.939	3.516	2.439	3.853	3.916	4.366
1.000000e-05	10	15	3.747	3.966	4.574	2.466	3.584	3.798	2.675
1.000000e-05	15	20	3.725	3.978	4.259	2.478	3.728	3.900	4.434
1.000000e-05	20	25	3.964	3.948	4.634	2.448	3.854	3.907	4.996
1.000000e-05	25	30	3.871	3.948	3.523	2.448	3.859	3.922	4.366

Short interpretation. Large time steps show explosive error growth (very large negative spatial slopes), consistent with explicit-scheme instability for diffusion-dominated dynamics. Within the stable subset, p_{L_2} ranges from 3.957 to 3.963, and p_{G_2} from 3.865 to 3.870.

9 Results: folder `example_2`

Stored mmsType: `example_2`.

Exact temperature used: $T_{\text{ex}} = A\cos(\beta\pi x)\cos(\beta\pi y)\sin(\beta\pi z)\sin(\beta\pi t)$ **Source term used:** $Q = A\cos(\beta\pi x)\cos(\beta\pi y)\sin(\beta\pi z)\left(\rho C_p\beta\pi\cos(\beta\pi t) + 3\alpha(\beta\pi)^2\sin(\beta\pi t)\right)$

Stored parameters (from `exactSolutionProperties.used`): $A = 1.0$, $\beta = 4.0$, $\gamma = 3.0$, $\omega = 4.0$.

Table 14: Available Δt values in `example_2` (descending order).

Δt
5.000000e-03
1.000000e-03
5.000000e-04
2.500000e-04
1.000000e-04
1.000000e-05

Short interpretation. Large time steps show explosive error growth (very large negative spatial slopes), consistent with explicit-scheme instability for diffusion-dominated dynamics. Within the stable subset, p_{L_2} ranges from 3.967 to 3.973, and p_{G_2} from 3.904 to 3.909.

Table 15: Global spatial fit rates p_{fit} (cell-centered norms) for `example_2`.

Δt	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	Comment
5.000000e-03	-54.556	-55.195	-56.005	-56.695	unstable/divergent
1.000000e-03	-151.896	-153.268	-154.400	-154.768	unstable/divergent
5.000000e-04	-71.521	-72.323	-73.049	-73.823	unstable/divergent
2.500000e-04	3.905	3.967	3.764	2.467	stable-like
1.000000e-04	3.909	3.971	3.763	2.471	stable-like
1.000000e-05	3.912	3.973	3.762	2.473	stable-like

Table 16: Global spatial fit rates p_{fit} (Gauss-point norms) for `example_2`.

Δt	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	-54.547	-55.214	-56.184
1.000000e-03	-151.766	-153.416	-154.926
5.000000e-04	-71.486	-72.436	-73.402
2.500000e-04	3.815	3.904	3.688
1.000000e-04	3.818	3.907	3.688
1.000000e-05	3.819	3.909	3.688

Table 17: Final errors on the finest mesh (typically $N = 30$) for `example_2`.

Δt	N	L_2	G_2	L_∞	G_∞	$error_T$	final $\|R\|_\infty$	steps	t_{end}
5.000000e-03	30	2.151270e+28	2.945364e+28	2.329873e+29	9.811018e+29	3.534898e+30	1.232871e+33	25	0.12500
1.000000e-03	30	3.854051e+72	5.403267e+72	4.560195e+73	2.018663e+74	6.332852e+74	2.440137e+77	125	0.12500
5.000000e-04	30	4.322776e+49	6.124657e+49	4.554453e+50	2.062396e+51	7.103045e+51	2.460392e+54	250	0.12500
2.500000e-04	30	1.743386e-03	2.282798e-03	8.003458e-03	2.343952e-02	2.864675e-01	1.101609e-02	500	0.12500
1.000000e-04	30	1.734826e-03	2.274754e-03	8.017566e-03	2.345021e-02	2.850610e-01	1.596621e-03	1250	0.12500
1.000000e-05	30	1.729723e-03	2.269959e-03	8.025995e-03	2.345662e-02	2.842225e-01	6.866831e-03	12501	0.12501

Table 18: Temporal fit slopes from finest-mesh errors: $e \sim (\Delta t)^{p_t}$ for `example_2`.

Metric	p_t (all available Δt)	p_t (stable-like subset)
L1	21.200	0.002
L2	21.459	0.002
Linf	21.637	-0.001
G1	21.155	0.002
G2	21.470	0.002
GInf	21.717	-0.000

Table 19: Pairwise spatial convergence rates (all available Δt) for `example_2`.

Δt	N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	10	15	-64.557	-65.488	-67.379	-66.988	-64.553	-65.262	-65.703
5.000000e-03	15	20	-52.505	-52.548	-51.179	-54.048	-52.492	-52.730	-53.512
5.000000e-03	20	25	-47.435	-48.348	-50.361	-49.848	-47.462	-48.467	-50.258
5.000000e-03	25	30	-45.258	-46.168	-47.517	-47.668	-45.175	-46.245	-47.648
1.000000e-03	10	15	3.975	4.116	4.357	2.616	3.812	4.007	3.881
1.000000e-03	15	20	-96.746	-100.139	-102.809	-101.639	-96.175	-100.496	-103.986
1.000000e-03	20	25	-375.164	-375.875	-376.409	-377.375	-375.157	-375.879	-376.377
1.000000e-03	25	30	-317.212	-318.669	-320.697	-320.169	-317.212	-318.628	-320.656
5.000000e-04	10	15	3.977	4.118	4.358	2.618	3.814	4.008	3.881
5.000000e-04	15	20	3.848	3.838	3.472	2.338	3.773	3.804	3.333
5.000000e-04	20	25	3.891	3.913	3.517	2.413	3.838	3.872	4.047
5.000000e-04	25	30	-650.331	-657.720	-662.913	-659.220	-649.243	-658.199	-665.238
2.500000e-04	10	15	3.978	4.119	4.359	2.619	3.814	4.009	3.880
2.500000e-04	15	20	3.851	3.842	3.475	2.342	3.775	3.806	3.333
2.500000e-04	20	25	3.902	3.924	3.509	2.424	3.845	3.880	4.047
2.500000e-04	25	30	3.850	3.959	3.348	2.459	3.868	3.919	3.410
1.000000e-04	10	15	3.978	4.119	4.359	2.619	3.815	4.009	3.880
1.000000e-04	15	20	3.854	3.844	3.476	2.344	3.777	3.808	3.333
1.000000e-04	20	25	3.909	3.930	3.504	2.430	3.850	3.884	4.047
1.000000e-04	25	30	3.864	3.973	3.340	2.473	3.877	3.929	3.408
1.000000e-05	10	15	3.979	4.120	4.360	2.620	3.815	4.010	3.880
1.000000e-05	15	20	3.855	3.846	3.477	2.346	3.778	3.809	3.333
1.000000e-05	20	25	3.913	3.933	3.501	2.433	3.852	3.887	4.047
1.000000e-05	25	30	3.872	3.982	3.335	2.482	3.883	3.935	3.406

10 Results: folder `example_3`

Stored mmsType: `example_3`.

Exact temperature used: $T_{\text{ex}} = A\cos(\beta\pi x)\cos(\beta\pi y)\cos(\beta\pi z)\sin(\beta\pi t)$ **Source term used:** $Q = A\cos(\beta\pi x)\cos(\beta\pi y)\cos(\beta\pi z)\left(\rho C_p\beta\pi\cos(\beta\pi t) + 3\alpha(\beta\pi)^2\sin(\beta\pi t)\right)$

Stored parameters (from `exactSolutionProperties.used`): $A = 1.0$, $\beta = 4.0$, $\gamma = 3.0$, $\omega = 4.0$.

Table 20: Available Δt values in `example_3` (descending order).

Δt
5.000000e-03
1.000000e-03
5.000000e-04
2.500000e-04
1.000000e-04
1.000000e-05

Short interpretation. Large time steps show explosive error growth (very large negative spatial slopes), consistent with explicit-scheme instability for diffusion-dominated dynamics. Within the stable subset, p_{L_2} ranges from 3.937 to 3.950, and p_{G_2} from 3.922 to 3.931.

11 Results: folder `example_4`

Stored mmsType: `example_4`.

Exact temperature used: $T_{\text{ex}} = A\sin(kx)\sin(ky)\sin(kz)\sin(\omega\pi t),\ k = \frac{1}{2}\gamma\pi$ **Source term used:** $Q = \rho C_p\,\partial_t T_{\text{ex}} - \alpha\nabla^2 T_{\text{ex}} = A\sin(kx)\sin(ky)\sin(kz)\left(\rho C_p\omega\pi\cos(\omega\pi t) + 3\alpha k^2\sin(\omega\pi t)\right)$

Table 21: Global spatial fit rates p_{fit} (cell-centered norms) for `example_3`.

Δt	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	Comment
5.000000e-03	-52.955	-53.008	-53.343	-54.508	unstable/divergent
1.000000e-03	-91.645	-91.672	-92.093	-93.172	unstable/divergent
5.000000e-04	3.886	3.937	3.704	2.437	stable-like
2.500000e-04	3.893	3.944	3.707	2.444	stable-like
1.000000e-04	3.897	3.948	3.708	2.448	stable-like
1.000000e-05	3.899	3.950	3.709	2.450	stable-like

Table 22: Global spatial fit rates p_{fit} (Gauss-point norms) for `example_3`.

Δt	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	-52.809	-52.824	-52.678
1.000000e-03	-91.351	-91.329	-91.036
5.000000e-04	3.814	3.922	4.446
2.500000e-04	3.819	3.927	4.444
1.000000e-04	3.821	3.929	4.443
1.000000e-05	3.823	3.931	4.442

Table 23: Final errors on the finest mesh (typically $N = 30$) for `example_3`.

Δt	N	L_2	G_2	L_∞	G_∞	$error_T$	final $\|R\|_\infty$	steps	t_{end}
5.000000e-03	30	1.491988e+24	1.492579e+24	6.417102e+24	8.785831e+24	2.451586e+26	2.474957e+28	25	0.12500
1.000000e-03	30	6.936892e+48	6.207645e+48	3.910339e+49	3.890913e+49	1.139848e+51	1.498366e+53	125	0.12500
5.000000e-04	30	1.849106e-03	2.353291e-03	8.058094e-03	1.113863e-02	3.038390e-01	3.087358e-02	250	0.12500
2.500000e-04	30	1.834616e-03	2.339671e-03	8.034973e-03	1.118092e-02	3.014581e-01	1.106320e-02	500	0.12500
1.000000e-04	30	1.825928e-03	2.331509e-03	8.021102e-03	1.120629e-02	3.000306e-01	1.928229e-03	1250	0.12500
1.000000e-05	30	1.820749e-03	2.326645e-03	8.012839e-03	1.122141e-02	2.991796e-01	7.108645e-03	12501	0.12501

Table 24: Temporal fit slopes from finest-mesh errors: $e \sim (\Delta t)^{p_t}$ for `example_3`.

Metric	p_t (all available Δt)	p_t (stable-like subset)
L1	14.335	0.004
L2	14.355	0.003
Linf	14.366	0.001
G1	14.284	0.003
G2	14.304	0.003
GInf	14.344	-0.002

Table 25: Pairwise spatial convergence rates (all available Δt) for `example_3`.

Δt	N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	10	15	-50.739	-50.818	-52.377	-52.318	-50.362	-50.220	-49.715
5.000000e-03	15	20	-57.638	-57.945	-58.297	-59.445	-57.718	-58.006	-58.364
5.000000e-03	20	25	-51.643	-51.467	-50.882	-52.967	-51.479	-51.457	-51.109
5.000000e-03	25	30	-48.960	-48.681	-47.227	-50.181	-48.885	-48.676	-48.570
1.000000e-03	10	15	3.646	3.813	3.127	2.313	3.632	3.907	3.981
1.000000e-03	15	20	4.200	4.014	3.898	2.514	3.967	3.956	4.758
1.000000e-03	20	25	-231.828	-231.677	-231.284	-233.177	-230.091	-230.195	-229.169
1.000000e-03	25	30	-358.428	-358.933	-361.042	-360.433	-358.625	-358.998	-360.936
5.000000e-04	10	15	3.648	3.814	3.127	2.314	3.634	3.908	3.981
5.000000e-04	15	20	4.207	4.022	3.901	2.522	3.972	3.962	4.761
5.000000e-04	20	25	3.835	3.988	4.678	2.488	3.838	3.903	5.555
5.000000e-04	25	30	3.790	3.961	2.911	2.461	3.854	3.888	2.731
2.500000e-04	10	15	3.649	3.815	3.127	2.315	3.634	3.909	3.981
2.500000e-04	15	20	4.211	4.025	3.903	2.525	3.974	3.965	4.763
2.500000e-04	20	25	3.846	3.998	4.681	2.498	3.845	3.910	5.557
2.500000e-04	25	30	3.812	3.983	2.920	2.483	3.869	3.904	2.705
1.000000e-04	10	15	3.649	3.816	3.127	2.316	3.635	3.909	3.981
1.000000e-04	15	20	4.213	4.028	3.904	2.528	3.976	3.966	4.763
1.000000e-04	20	25	3.853	4.004	4.682	2.504	3.850	3.914	5.558
1.000000e-04	25	30	3.825	3.996	2.925	2.496	3.879	3.913	2.689
1.000000e-05	10	15	3.649	3.816	3.127	2.316	3.635	3.910	3.981
1.000000e-05	15	20	4.214	4.029	3.904	2.529	3.977	3.968	4.764
1.000000e-05	20	25	3.857	4.008	4.683	2.508	3.852	3.917	5.558
1.000000e-05	25	30	3.833	4.004	2.928	2.504	3.885	3.919	2.680

Stored parameters (from `exactSolutionProperties.used`): $A = 1.0$, $\beta = 4.0$, $\gamma = 3.0$, $\omega = 4.0$.

Table 26: Available Δt values in `example_4` (descending order).

Δt
5.000000e-03
1.000000e-03
5.000000e-04
2.500000e-04
1.000000e-04
1.000000e-05

Short interpretation. Large time steps show explosive error growth (very large negative spatial slopes), consistent with explicit-scheme instability for diffusion-dominated dynamics. Within the stable subset, p_{L_2} ranges from 3.223 to 4.951, and p_{G_2} from 3.355 to 4.643.

12 Global Discussion

- Across examples, large time steps generally lead to unstable/divergent behavior, reflected by large negative fitted spatial slopes.
- For stable-like subsets, observed spatial orders are approximately fourth order in L_2 and around fourth order in G_2, depending on the MMS case.
- This version explicitly includes pairwise spatial convergence tables for every example and every available Δt.

Table 27: Global spatial fit rates p_{fit} (cell-centered norms) for `example_4`.

Δt	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	Comment
5.000000e-03	-48.933	-49.811	-50.755	-51.311	unstable/divergent
1.000000e-03	-206.070	-208.549	-210.728	-210.049	unstable/divergent
5.000000e-04	-180.113	-182.324	-184.172	-183.824	unstable/divergent
2.500000e-04	3.193	3.223	3.372	1.723	stable-like
1.000000e-04	4.052	4.067	3.921	2.567	stable-like
1.000000e-05	5.159	4.951	3.548	3.451	stable-like

Table 28: Global spatial fit rates p_{fit} (Gauss-point norms) for `example_4`.

Δt	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	-48.922	-49.840	-50.824
1.000000e-03	-205.936	-208.697	-210.999
5.000000e-04	-179.956	-182.428	-184.332
2.500000e-04	3.337	3.355	4.098
1.000000e-04	4.000	4.030	3.820
1.000000e-05	4.654	4.643	3.679

Table 29: Final errors on the finest mesh (typically $N = 30$) for `example_4`.

Δt	N	L_2	G_2	L_∞	G_∞	$error_T$	final $\|R\|_\infty$	steps	t_{end}
5.000000e-03	30	1.241005e+30	1.752587e+30	3.384822e+31	1.172736e+32	2.039180e+32	2.473231e+35	25	0.12500
1.000000e-03	30	1.809660e+91	2.564984e+91	5.306231e+92	1.834765e+93	2.973575e+93	3.875946e+96	125	0.12500
5.000000e-04	30	1.958097e+97	2.779876e+97	5.126847e+98	1.777951e+99	3.217482e+99	3.748419e+102	250	0.12500
2.500000e-04	30	9.715900e-05	1.073237e-04	3.240713e-04	3.865856e-04	1.596485e-02	1.169685e-02	500	0.12500
1.000000e-04	30	3.773190e-05	4.933325e-05	1.726853e-04	5.406700e-04	6.199984e-03	3.619999e-04	1250	0.12500
1.000000e-05	30	1.323858e-05	2.334927e-05	2.678560e-04	6.393165e-04	2.175320e-03	4.889751e-03	12501	0.12501

Table 30: Temporal fit slopes from finest-mesh errors: $e \sim (\Delta t)^{p_t}$ for `example_4`.

Metric	p_t (all available Δt)	p_t (stable-like subset)
L1	27.467	0.689
L2	27.890	0.587
Linf	28.038	0.011
G1	27.350	0.459
G2	27.863	0.445
GInf	28.107	-0.140

Table 31: Pairwise spatial convergence rates (all available Δt) for `example_4`.

Δt	N_1	N_2	p_{L_1}	p_{L_2}	p_{L_∞}	p_{error_T}	p_{G_1}	p_{G_2}	p_{G_∞}
5.000000e-03	10	15	-54.079	-55.211	-56.420	-56.711	-54.058	-55.250	-56.489
5.000000e-03	15	20	-47.725	-48.451	-49.280	-49.951	-47.730	-48.479	-49.391
5.000000e-03	20	25	-45.434	-46.168	-46.951	-47.668	-45.410	-46.189	-46.985
5.000000e-03	25	30	-44.248	-45.094	-45.901	-46.594	-44.242	-45.109	-45.934
1.000000e-03	10	15	3.331	3.414	3.125	1.914	3.463	3.526	3.951
1.000000e-03	15	20	-308.315	-314.834	-319.957	-316.334	-308.039	-315.406	-321.647
1.000000e-03	20	25	-326.511	-327.696	-328.991	-329.196	-326.507	-327.702	-329.011
1.000000e-03	25	30	-292.323	-293.565	-294.794	-295.065	-292.323	-293.568	-294.797
5.000000e-04	10	15	3.844	3.916	3.468	2.416	3.847	3.917	3.903
5.000000e-04	15	20	2.719	2.732	3.192	1.232	2.992	2.989	4.590
5.000000e-04	20	25	-423.935	-433.234	-440.739	-434.734	-423.638	-434.052	-444.050
5.000000e-04	25	30	-742.053	-743.074	-744.225	-744.574	-742.050	-743.080	-744.220
2.500000e-04	10	15	4.152	4.215	3.592	2.715	4.063	4.134	3.880
2.500000e-04	15	20	3.573	3.593	3.820	2.093	3.667	3.686	4.335
2.500000e-04	20	25	2.348	2.322	3.109	0.822	2.711	2.683	3.770
2.500000e-04	25	30	1.013	1.089	2.002	-0.411	1.540	1.493	4.719
1.000000e-04	10	15	4.362	4.409	3.537	2.909	4.203	4.269	3.866
1.000000e-04	15	20	4.362	4.355	4.265	2.855	4.203	4.222	4.116
1.000000e-04	20	25	3.892	3.863	4.148	2.363	3.891	3.898	3.359
1.000000e-04	25	30	2.599	2.675	3.627	1.175	3.060	3.065	3.771
1.000000e-05	10	15	4.496	4.530	3.511	3.030	4.289	4.351	3.858
1.000000e-05	15	20	5.028	4.907	4.178	3.407	4.578	4.574	3.993
1.000000e-05	20	25	5.767	5.481	3.177	3.981	4.970	4.913	3.152
1.000000e-05	25	30	6.251	5.251	2.633	3.751	5.304	5.153	3.325

13 Conclusions

The report consolidates all currently available examples (0 through 4) into a traceable MMS summary for the explicit solver, with both fitted and pairwise spatial rates extracted directly from the latest `results/` dataset.

Referenced Files

- Solver source: `/home/pablo/cardiacFoam-pc/applications/solvers/testHighOrderHeatEquationExpli`
- Run script: `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquation`
- Results root: `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquati`